CLOUD COMPUTING MODULE | WEEK 1

Azure Virtual Machine & Azure File Share

Assignment Report: requirements, solution, evidence and lessons learned

Students	Waseem Isaac & Dana Obid
Module	Cloud Module, Week 1
Platform	Microsoft Azure (Free Trial subscription)
Azure region	Sweden Central
Operating system	Ubuntu Server 22.04 LTS (Canonical)
Report date	November 6, 2026

All screenshots in this report were taken during the assignment. Personal identifiers (e-mail, subscription ID, directory name) are redacted; secrets such as keys are never shown.

1. Summary

This report documents the complete solution of the Week 1 assignment: creating and managing a virtual machine on Microsoft Azure, connecting to it, and creating and using an Azure File Share from both the virtual machine (Linux) and a personal Windows PC. The optional tasks on Linux were also completed: an Apache web site, a MySQL database and a Samba file share.

Every step below follows the same pattern: what the requirement asks for, how it was solved (portal actions or commands), and the screenshot that proves it.

1.1 Requirements overview

#	Requirement	Solution	Status
1	Create an Azure account	Azure Free Trial subscription; plan and spending limit verified with Azure CLI	**Completed**
2	Create a resource group	WaseemIsaac-RG	**Completed**
3	Create a virtual machine	WaseemIsaac-VM: Ubuntu Server 22.04 LTS, Standard_B2ats_v2, Sweden Central, SSH key authentication	**Completed**
4	Connect to the virtual machine	SSH from Azure Cloud Shell to the public IP address	**Completed**
5	Create and connect an Azure File Share	Storage account waseemisaac2026wa, share myshare (32 GiB); mounted with CIFS on Linux and mapped as drive Z: on Windows	**Completed**
6	Optional tasks on the VM (Linux)	Apache web site, MySQL database, Samba share, and create/read/delete files on the Azure File Share	**Completed**
7	Submission (GitHub and LinkedIn)	Public repository with this report and a LinkedIn post (links in Section 10)	**To do**
8	Clean up resources	Delete the resource group after submission (command in Section 9.2)	**To do**

1.2 Environment at a glance

Item	Value
Subscription type	Azure Free Trial (spending limit: On)
Resource group	WaseemIsaac-RG
Virtual machine	WaseemIsaac-VM
Region (VM and storage)	Sweden Central
Image	Canonical Ubuntu Server 22.04 LTS, x64 Gen2
Size	Standard_B2ats_v2 (2 vCPU, 1 GiB memory)
Public IP address	20.240.53.172
Storage account	waseemisaac2026wa (locally redundant)

Item	Value
File share	myshare, SMB, 32 GiB provisioned

2. Step 1: Create an Azure Account

Requirement: create a free Azure trial account (student benefits can be used).

Solution

1. Open azure.microsoft.com/free and sign in with a Microsoft account.
2. Complete the identity verification (phone number and payment card, used for verification only).
3. Open the Azure portal and confirm that the active subscription is the free trial one.

Because the account was created with a work e-mail address, the subscription type was verified explicitly from Azure Cloud Shell instead of trusting the portal labels:

```
az rest --method get \
  --url "https://management.azure.com/subscriptions/<SUBSCRIPTION_ID>?api-version=2020-01-01" \
  --query "{plan:subscriptionPolicies.quotaId, spendingLimit:subscriptionPolicies.spendingLimit}"

# Output
{
  "plan": "FreeTrial_2014-09-01",
  "spendingLimit": "On"
}
```

> **Result:** the plan is FreeTrial and the spending limit is On, so services stop instead of being billed when the trial credit ends. The Upgrade button was deliberately not used, to keep this protection.

3. Step 2: Create a Resource Group

Requirement: create a resource group with a meaningful name and a location.

Solution

1. Azure portal > Resource groups > Create.
2. Subscription: Azure subscription 1. Resource group name: WaseemIsaac-RG. Region: West Europe.
3. Select Review + create, then Create.

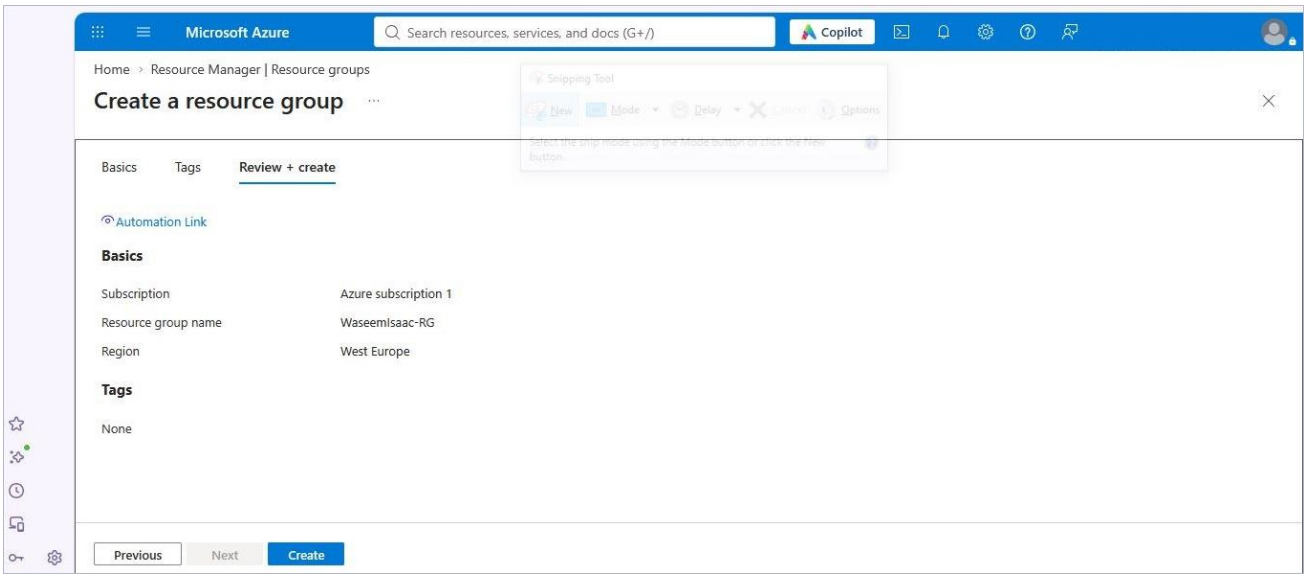


Figure 1. *Review + create page of the resource group WaseemIsaac-RG (account e-mail redacted).*

The resource group region only stores the group's metadata. The resources inside it (virtual machine and storage) are deployed in Sweden Central, which is allowed by Azure.

4. Step 3: Create the Virtual Machine

Requirement: create a Linux or Windows VM with a suitable size, a public IP address and an SSH/RDP rule.

4.1 Chosen configuration

Setting	Value	Reason
Resource group	WaseemIsaac-RG	Keeps every resource together for easy clean-up
VM name	WaseemIsaac-VM	Hyphens are accepted; underscores are not
Region	Sweden Central	Region where the chosen size is allowed for this subscription
Image	Canonical Ubuntu Server 22.04 LTS	Official image; no paid Marketplace offer needed
Size	Standard_B2ats_v2	Burstable size available for the free trial
OS disk	Standard SSD (LRS)	Cheaper than Premium SSD
Authentication	SSH key, user azureuser	More secure than a password
Networking	New VNet, Standard public IP, NSG rule SSH (22)	Required for remote access

4.2 Creation command

The VM was created from Azure Cloud Shell with the Azure CLI, because the portal wizard repeatedly selected a third-party Marketplace image that a trial subscription cannot purchase (see Section 8). The command pins the official Canonical image explicitly:

```
az vm create \
  --resource-group WaseemIsaac-RG \
  --name WaseemIsaac-VM \
  --location swedencentral \
  --image Canonical:0001-com-ubuntu-server-jammy:22_04-lts-gen2:latest \
  --size Standard_B2ats_v2 \
  --storage-sku StandardSSD_LRS \
  --admin-username azureuser \
  --generate-ssh-keys \
  --public-ip-sku Standard \
  --nsg-rule SSH

# Output (excerpt)
"location": "swedencentral",
"powerState": "VM running",
"privateIpAddress": "10.0.0.4",
"publicIpAddress": "20.240.53.172",
"resourceGroup": "WaseemIsaac-RG"
```

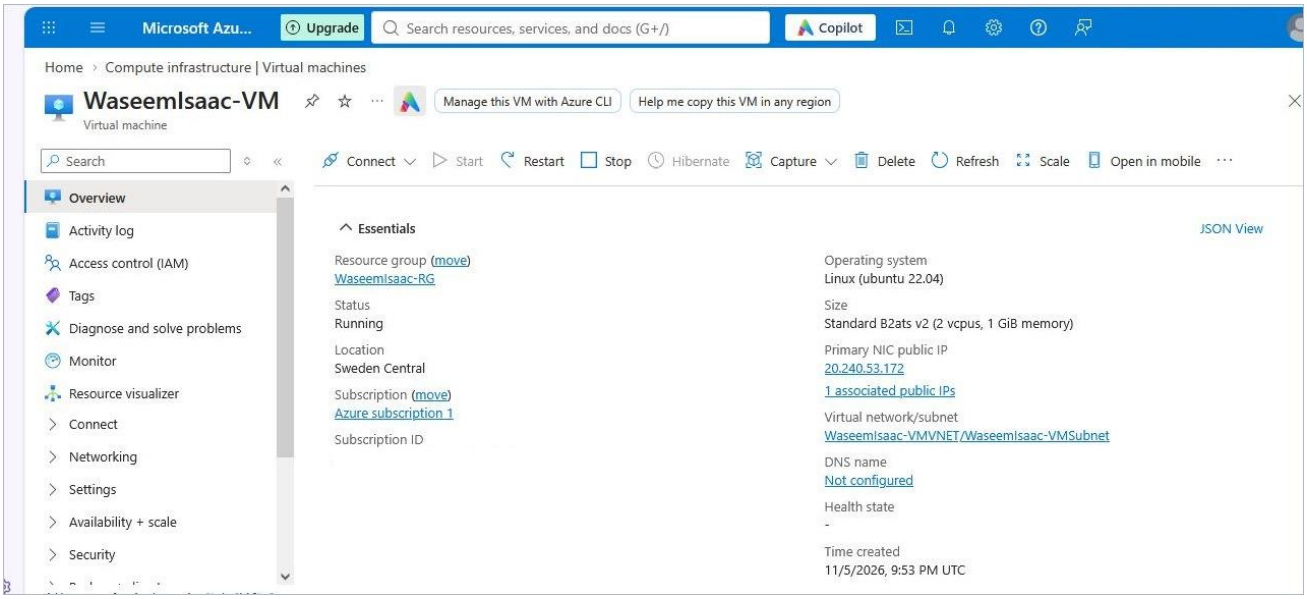


Figure 2. *Virtual machine overview: Running, Sweden Central, Standard B2ats v2 (2 vCPUs, 1 GiB), public IP 20.240.53.172 (e-mail and subscription ID redacted).*

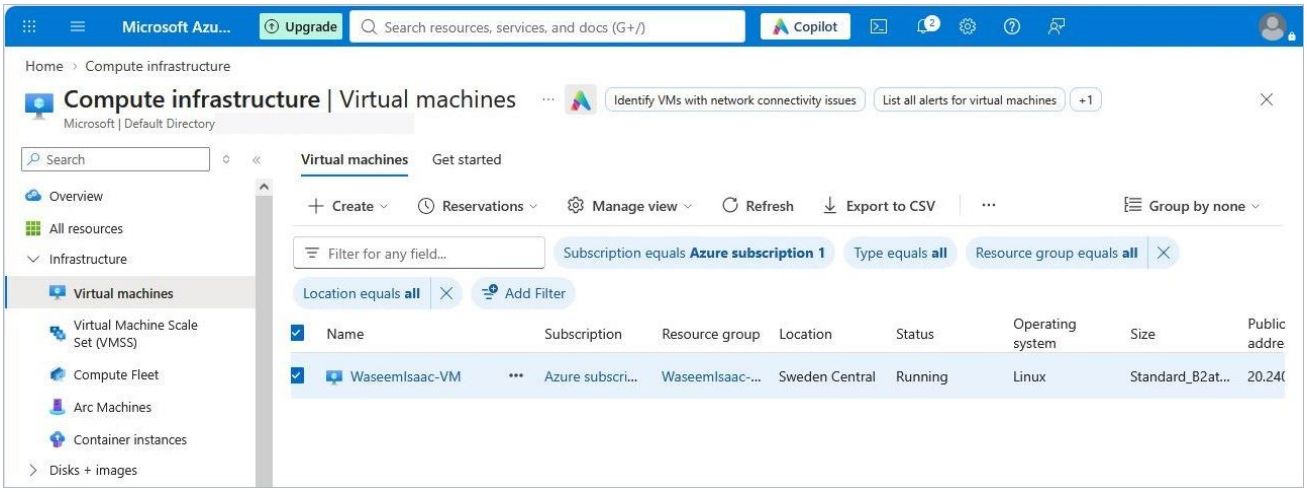


Figure 3. *Virtual machines list showing WaseemIsaac-VM in the Running state (e-mail and directory name redacted).*

5. Step 4: Connect to the Virtual Machine

Requirement: connect over SSH (Linux) with the public IP address.

Solution

1. Copy the public IP address (20.240.53.172) from the VM overview page.
2. Open Azure Cloud Shell (Bash) and run the SSH command below.
3. Type yes to accept the host fingerprint on the first connection.

```
ssh azureuser@20.240.53.172
```

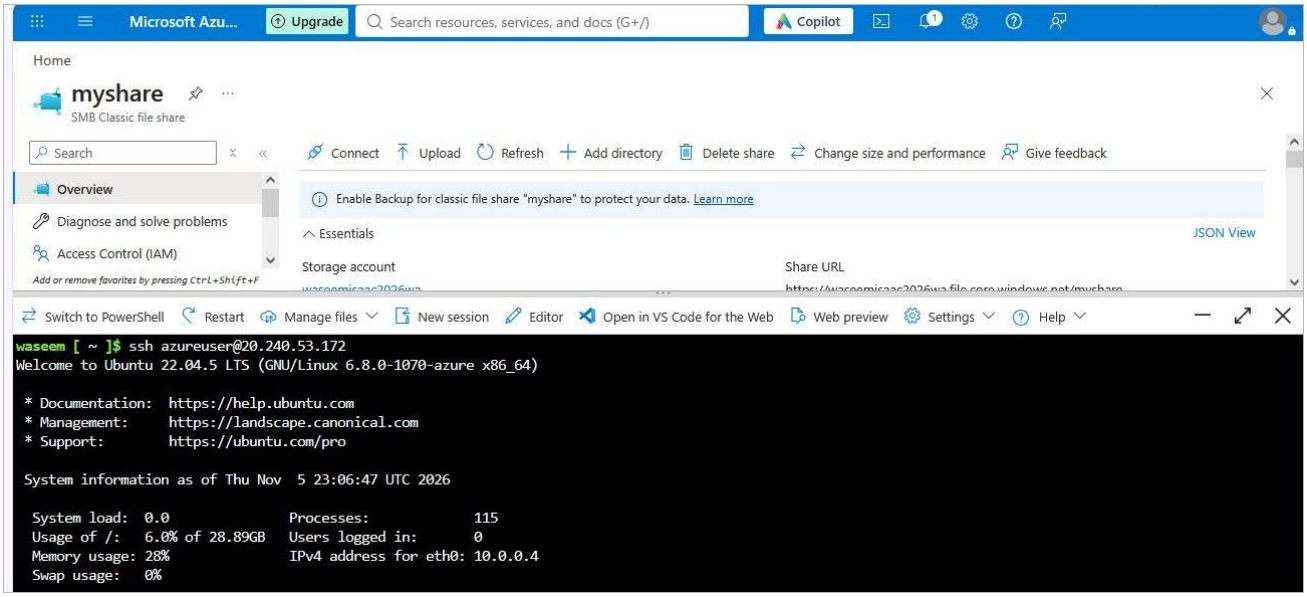


Figure 4. *Successful SSH session from Azure Cloud Shell: the Ubuntu 22.04.5 LTS welcome banner and system information (e-mail redacted).*

> **Note:** the Ubuntu image disables password logins, so authentication works only with the SSH key. The key was kept private and is not part of this report.

6. Step 5: Create and Connect an Azure File Share

Requirement: create a storage account and a file share, then connect it to the virtual machine (CIFS) and to the PC (mapped drive).

6.1 Create the storage account

1. Azure portal > Storage accounts > Create.
2. Resource group: WaseemIsaac-RG. Storage account name: waseemisaac2026wa (lowercase letters and digits, globally unique).
3. Region: Sweden Central, so it sits next to the VM. Redundancy: locally redundant (LRS), the cheapest option.
4. Review + create > Create and wait for “Your deployment is complete”.

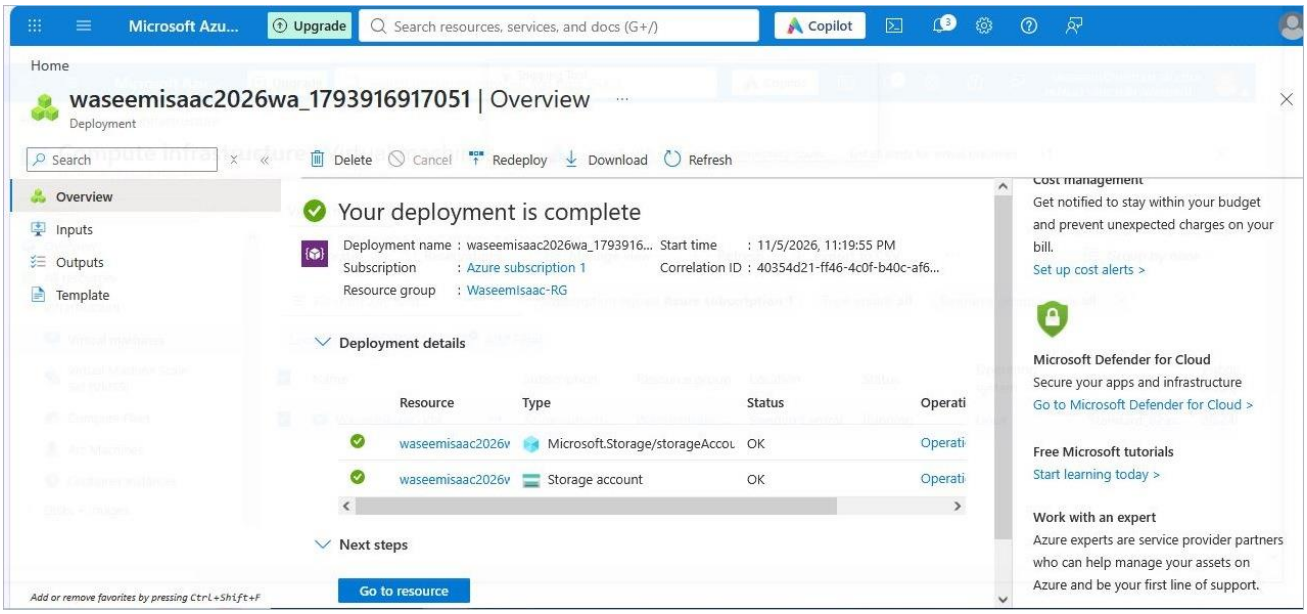


Figure 5. *Deployment of the storage account waseemisaac2026wa completed in WaseemIsaac-RG (e-mail redacted).*

6.2 Create the file share

1. Open the storage account > Data storage > File shares > + File share.
2. Name: myshare. Provisioned storage: 32 GiB. Protocol: SMB.
3. Review + create > Create.

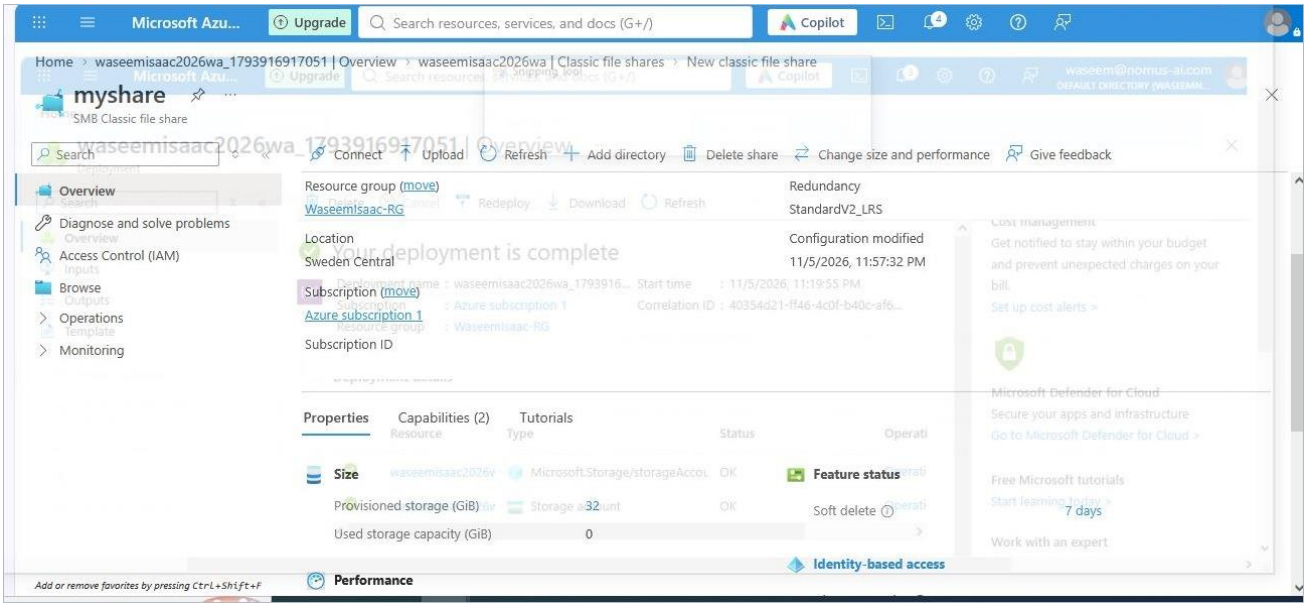


Figure 6. *The file share myshare: SMB, Sweden Central, redundancy StandardV2_LRS, 32 GiB provisioned (e-mail and subscription ID redacted).*

6.3 Mount the share on the Linux VM (CIFS)

The storage account key is available under Security + networking > Access keys. It is stored in a shell variable so it is not typed in the command itself. It must never be published.

```
sudo apt update && sudo apt install -y cifs-utils
sudo mkdir -p /mnt/myshare

STORAGE_KEY='<storage-account-key>'
sudo mount -t cifs //waseemisaac2026wa.file.core.windows.net/myshare /mnt/myshare \
  -o
username=waseemisaac2026wa,password="$STORAGE_KEY",dir_mode=0777,file_mode=0777,serverino,nosharesock
,actimeo=30
```

Create, read and delete test (the three file operations required by the assignment):

```
df -h | grep myshare                       # confirm the mount (32G share)
echo "hello azure" > /mnt/myshare/test.txt # create
cat /mnt/myshare/test.txt                  # read
rm /mnt/myshare/test.txt                   # delete
```

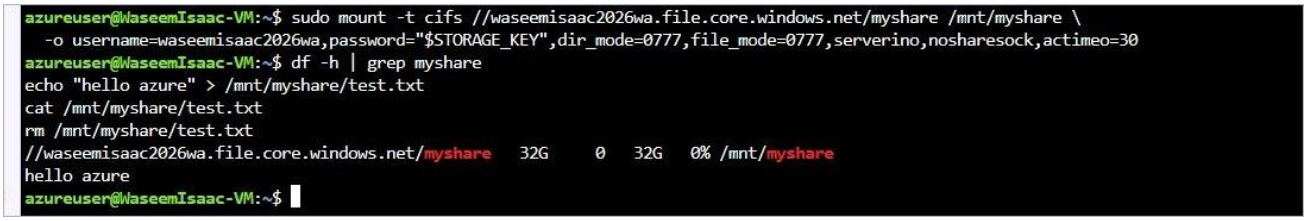


Figure 7. *The share is mounted on /mnt/myshare (32G) and the file was created and read back ("hello azure").*

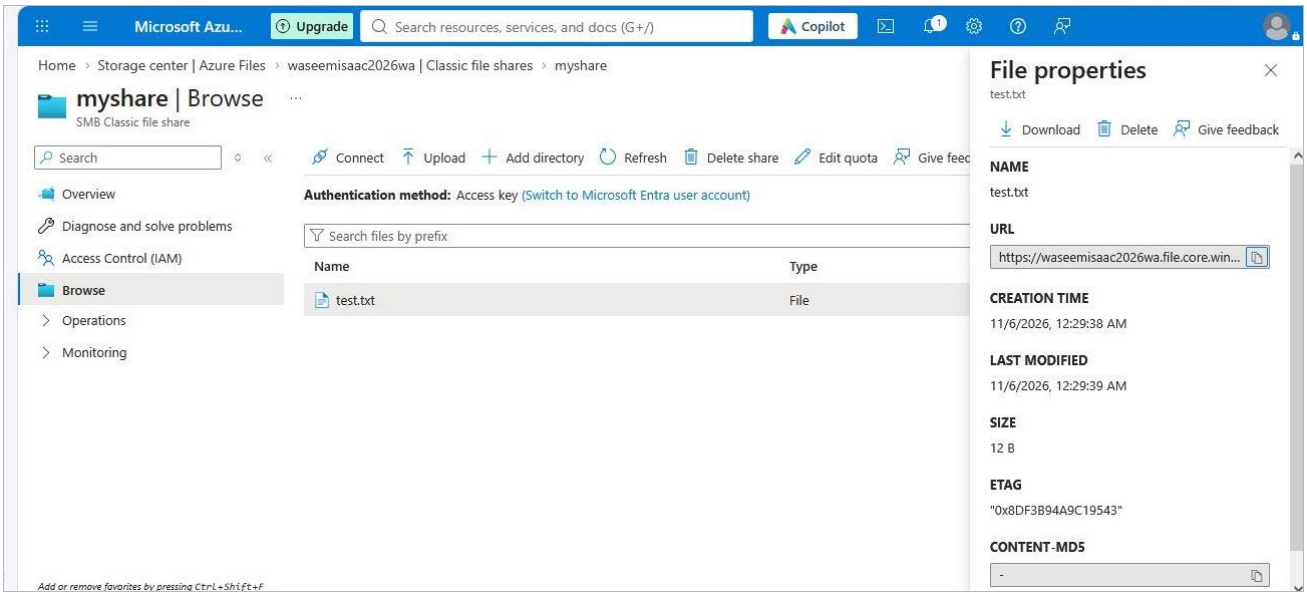


Figure 8. *Portal Browse view of myshare: the file test.txt (12 bytes) written from the VM is visible in Azure (e-mail redacted).*

6.4 Map the share on the Windows PC

From an elevated PowerShell window on the PC, the share was mapped as drive Z:. TCP port 445 was reachable from this network. The key was entered at the password prompt.

```
$key = "<storage-account-key>"
net use Z: \\waseemisaac2026wa.file.core.windows.net\myshare /user:localhost\waseemisaac2026wa $key
```



Figure 9. *PowerShell: "The command completed successfully." after mapping drive Z:.*

Create, list and read a file on the mapped drive:

```
Set-Content Z:\test-from-pc.txt "hello from my PC"
Get-ChildItem Z:\
Get-Content Z:\test-from-pc.txt
```

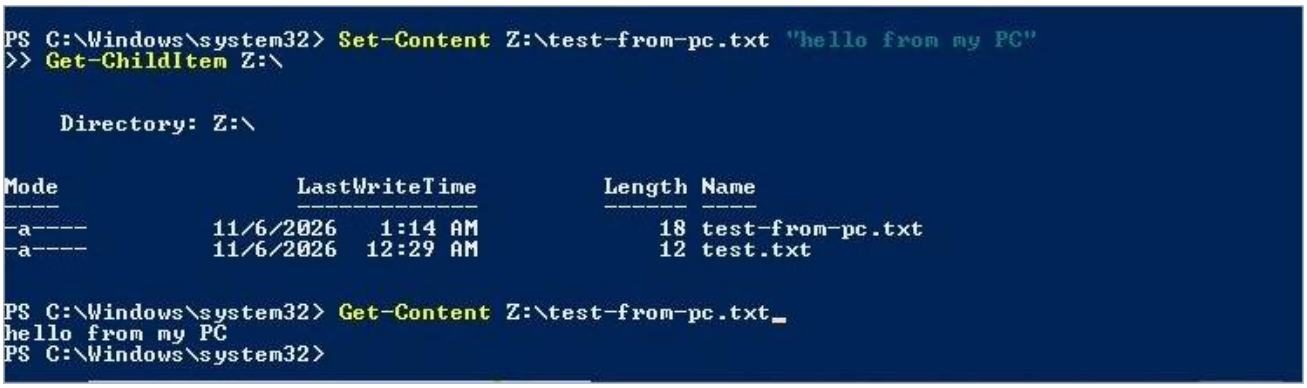


Figure 10. *Windows PC: test-from-pc.txt created on Z: and read back ("hello from my PC").*

6.5 Proof that both machines use the same share

A file created from the PC (shared-check.txt) was then opened on the Linux VM, which proves that the PC and the VM are connected to the same Azure File Share.

```
ls /mnt/myshare
cat /mnt/myshare/shared-check.txt
```

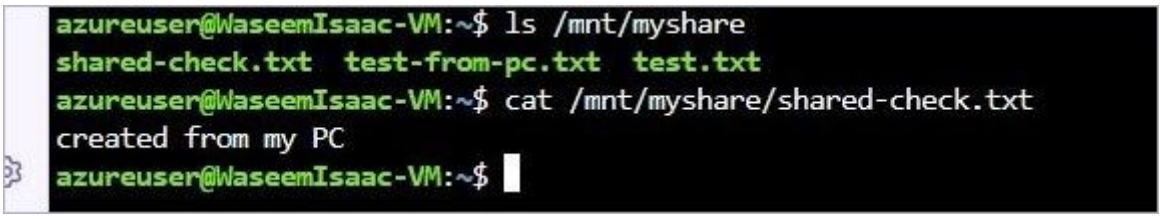


Figure 11. *On the VM: shared-check.txt (created from the PC) is listed and its content "created from my PC" is displayed.*

6.6 File operations summary

Operation	From	Command	Evidence
Create	Linux VM	echo ... > /mnt/myshare/test.txt	Figures 7 and 8
Read	Linux VM	cat /mnt/myshare/test.txt	Figure 7
Delete	Linux VM	rm /mnt/myshare/test.txt	Figure 7
Create and read	Windows PC	Set-Content / Get-Content on Z:	Figures 9 and 10
Cross-check	PC to VM	cat /mnt/myshare/shared-check.txt	Figure 11

7. Step 6: Optional Tasks on the Linux VM

7.1 Add swap space (1 GiB RAM)

The chosen size has only 1 GiB of memory, so a 1 GiB swap file was added before installing the database server, to avoid out-of-memory failures.

```
sudo fallocate -l 1G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
free -h
```

```
azureuser@WaseemIsaac-VM:~$ sudo fallocate -l 1G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
free -h
Setting up swapspace version 1, size = 1024 MiB (1073737728 bytes)
no label, UUID=f05a4bd2-8c7a-48a2-924d-cd6eb25db49a
               total        used        free      shared  buff/cache   available
Mem:           891Mi       242Mi       126Mi       3.0Mi       522Mi       500Mi
Swap:          1.0Gi          0B       1.0Gi
azureuser@WaseemIsaac-VM:~$
```

Figure 12. *Swap space of 1.0 GiB active (free -h).*

7.2 Apache web server and a web site

```
sudo apt install -y apache2
echo "<h1>Hello from Azure - Waseem Isaac</h1>" | sudo tee /var/www/html/index.html
sudo systemctl status apache2 --no-pager | head -5
```

Port 80 was opened in the network security group from Azure Cloud Shell (not from inside the VM, where the Azure CLI is not installed):

```
az vm open-port -g WaseemIsaac-RG -n WaseemIsaac-VM --port 80 --priority 1010
```

```
Setting up swapspace version 1, size = 1024 MiB (1073737728 bytes)
no label, UUID=f05a4bd2-8c7a-48a2-924d-cd6eb25db49a
               total        used        free      shared  buff/cache   available
Mem:           891Mi       242Mi       126Mi       3.0Mi       522Mi       500Mi
Swap:          1.0Gi          0B       1.0Gi
azureuser@WaseemIsaac-VM:~$ sudo apt install -y apache2
echo "<h1>Hello from Azure - Waseem Isaac</h1>" | sudo tee /var/www/html/index.html
sudo systemctl status apache2 --no-pager | head -5
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap mime-support ssl-cert
```

Figure 13. *Installation of Apache2 and creation of the home page on the VM.*



Figure 14. *The published web site opened from outside the VM at http://20.240.53.172.*

7.3 MySQL database server

```
sudo apt install -y mysql-server
sudo mysql -e "CREATE DATABASE studentdb; SHOW DATABASES;"
```

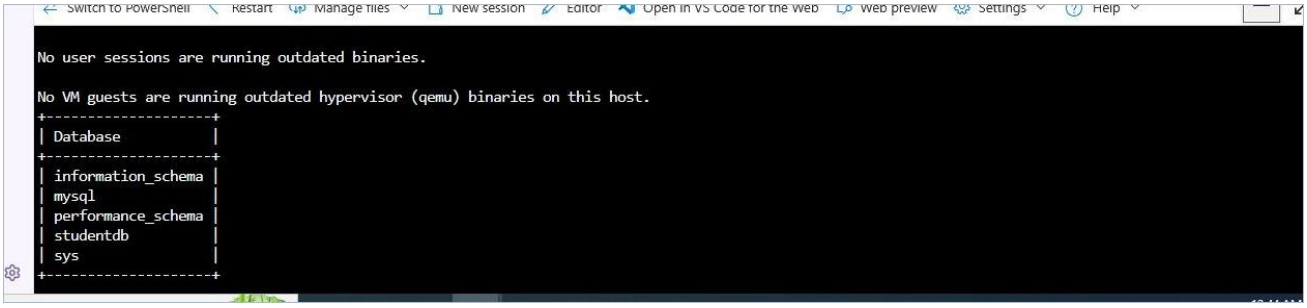


Figure 15. *SHOW DATABASES lists the newly created database studentdb.*

7.4 Samba file sharing

```
sudo apt install -y samba
sudo mkdir -p /srv/samba/share && sudo chmod 777 /srv/samba/share

# appended to /etc/samba/smb.conf
[share]
   path = /srv/samba/share
   read only = no
   browsable = yes

sudo smbpasswd -a azureuser
sudo systemctl restart smbd
sudo systemctl status smbd --no-pager | head -5
```

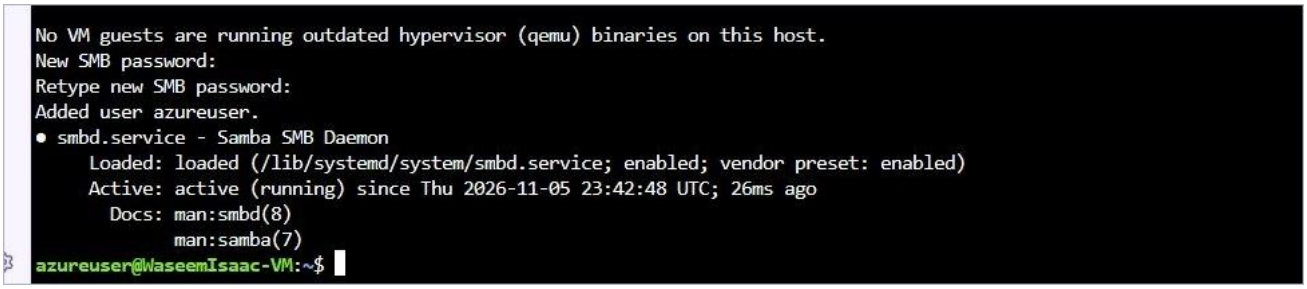


Figure 16. *Samba user azureuser added and the smbd service is active (running).*

The share was tested locally with smbclient. The port 445 was intentionally not opened to the Internet, to avoid exposing Samba publicly.

```
sudo apt install -y smbclient
smbclient //localhost/share -U azureuser -c "ls"
```

```
Preparing to unpack .../libsmbclient_2%3a4.15.13+dfsg-0ubuntu1.13_amd64.deb ...
Unpacking libsmbclient:amd64 (2:4.15.13+dfsg-0ubuntu1.13) ...
Selecting previously unselected package smbclient.
Preparing to unpack .../smbclient_2%3a4.15.13+dfsg-0ubuntu1.13_amd64.deb ...
Unpacking smbclient (2:4.15.13+dfsg-0ubuntu1.13) ...
Setting up libsmbclient:amd64 (2:4.15.13+dfsg-0ubuntu1.13) ...
Setting up smbclient (2:4.15.13+dfsg-0ubuntu1.13) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Password for [WORKGROUP\azureuser]:
  .                                   D        0  Thu Nov  5 23:41:49 2026
  ..                                  D        0  Thu Nov  5 23:41:49 2026

                30298176 blocks of size 1024. 26552392 blocks available
```

Figure 17. *smbclient connects to the Samba share and lists the (empty) directory.*

Create, read and delete through Samba:

```
echo "hello samba" > /tmp/samba-test.txt
smbclient //localhost/share -U azureuser -c "put /tmp/samba-test.txt samba-test.txt; ls; get samba-test.txt /tmp/samba-copy.txt; del samba-test.txt; ls"
cat /tmp/samba-copy.txt
```

```
azureuser@WaseemIsaac-VM:~$ echo "hello samba" > /tmp/samba-test.txt
smbclient //localhost/share -U azureuser -c "put /tmp/samba-test.txt samba-test.txt; ls; get samba-test.txt /tmp/samba-copy.txt; del samba-test.txt; ls"
cat /tmp/samba-copy.txt
Password for [WORKGROUP\azureuser]:
putting file /tmp/samba-test.txt as \samba-test.txt (5.9 kb/s) (average 5.9 kb/s)
  .                                   D        0  Thu Nov  5 23:51:52 2026
  ..                                  D        0  Thu Nov  5 23:41:49 2026
  samba-test.txt                      A       12  Thu Nov  5 23:51:52 2026

                30298176 blocks of size 1024. 26552384 blocks available
getting file \samba-test.txt of size 12 as /tmp/samba-copy.txt (120000.0 KiloBytes/sec) (average inf KiloBytes/sec)
  .                                   D        0  Thu Nov  5 23:51:52 2026
  ..                                  D        0  Thu Nov  5 23:41:49 2026

                30298176 blocks of size 1024. 26552384 blocks available
hello samba
azureuser@WaseemIsaac-VM:~$
```

Figure 18. *File uploaded (put), listed, downloaded (get) and deleted (del) over Samba; the copy contains “hello samba”.*

8. Problems Encountered and Solutions

#	Problem	Cause	Solution
1	Size Standard_B1s was “not available” (NotAvailableForSubscription) in North Europe, West Europe and Central US.	The size is restricted for this subscription in those regions.	Listed the sizes allowed per region with az vm list-skus and picked Standard_B2ats_v2 in Sweden Central.

#	Problem	Cause	Solution
2	It was unclear whether the account (work e-mail) was a free subscription.	The portal did not show the Offer column.	Queried the subscription policy with az rest: FreeTrial plan, spending limit On.
3	VM creation failed with BadRequest: the offer cannot be used to purchase paid Marketplace offers.	The portal selected a third-party Ubuntu image (not from Canonical); trial subscriptions cannot buy paid Marketplace images.	Created the VM with az vm create and the official Canonical image.
4	SSH failed with “Permission denied (publickey)”.	The VM accepts only the SSH key it was created with, and the client did not have it.	Generated a key pair in Cloud Shell and added the public key with az vm user update.
5	“az: command not found” on the VM.	The Azure CLI exists in Cloud Shell, not on the VM.	Ran az commands (for example open-port) from Cloud Shell and the Linux commands on the VM.
6	net use on Windows returned System error 86 (password not correct).	The storage key was not accepted when pasted into the interactive prompt.	Cleared old connections, then typed the key again at the prompt; the drive was mapped.
7	smbclient returned NT_STATUS_LOGON_FAILURE.	The Samba password differs from the Linux password and had been mistyped.	Reset it with sudo smbpasswd -a azureuser and tested again.
8	Only 1 GiB of RAM for MySQL.	Small burstable VM size.	Added a 1 GiB swap file before installing the database server.

9. Security, Cost Control and Clean-up

9.1 Good practices applied

- SSH key authentication only; the private key and the storage account key are not published.
- Samba port 445 was not exposed to the Internet; only ports 22 (SSH) and 80 (HTTP) were opened.
- Standard SSD, LRS replication and the smallest suitable VM size were chosen to minimize cost.
- The trial spending limit was kept enabled (no upgrade to pay-as-you-go).
- E-mail addresses, subscription ID and directory name were redacted from all screenshots in this report.

9.2 Clean up resources

After this report has been submitted, every resource is removed by deleting the resource group, which deletes the VM, disk, network, public IP and storage account together:

```
az group delete --name WaseemIsaac-RG --yes --no-wait
```

> **Reminder:** Cloud Shell may also have created its own storage account in a separate resource group (named like cloud-shell-storage-...). Delete it too if it is no longer needed.

10. Submission Details

Item	Value
Students names	Waseem Isaac & Dana Obid
GitHub repository (public)	github.com/waseemisaac91/Cloud-Azure-platform-assignment
LinkedIn post	https://lnkd.in/p/egCaad2K
Trainer (LinkedIn)	Yavuz Koksal: linkedin.com/in/yavuz-koksal

11. Conclusion

The assignment objectives were achieved: a Linux virtual machine was created and managed on Azure, accessed over SSH, and connected to an Azure File Share from both Linux (CIFS) and Windows (mapped drive), with files created, read and deleted from both sides. The optional Linux tasks (Apache, MySQL and Samba) were completed and verified. The main lessons were how subscription restrictions affect available VM sizes and images, why the Azure CLI is a reliable fallback when the portal wizard fails, and how important it is to protect keys and clean up resources.